Subham Agrawal

B.Tech, Civil and Infrastructure Engineering
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SUMMARY

Final-year IIT Jodhpur undergraduate with a minor in Mathematical and Computational Economics. Researches Indian market volatility in Python: India VIX, realised-volatility forecasting and option-hedging backtests, all tested out of sample. Seeking quant trading and research roles.

EDUCATION

Degree	Institute	Year
B.Tech, Civil and Infrastructure Engineering	Indian Institute of Technology Jodhpur	2023 – 2027
Minor, Mathematical and Computational Economics	Indian Institute of Technology Jodhpur	2026 – 2027

Coursework: Probability, Statistics and Stochastic Processes; Econometrics Lab; Games and Economic Behavior (Grade A); DSA; *ongoing:* Financial Derivatives and Risk Management, ML for Economics, Intro to Machine Learning

PROJECTS

Volatility Dynamics and the Fear Index: Nifty 50 and India VIX 2026
Team project, Financial Derivatives and Risk Management | *Python, pandas, statsmodels* Live app | GitHub

- Analysed 1,536 days of Nifty 50 and India VIX (2020–26): implied volatility exceeded realised on 78% of days (Newey-West $t = 2.8$), a persistent variance risk premium.
- Re-implemented NSE's VIX formula from option prices and split the market into low, normal and high fear regimes, including the June 2024 election shock.
- Backtested Black-Scholes-priced protective puts: a VIX-timed hedge cut maximum drawdown from −38% to −22%, most of it in the March 2020 crash.

Landslide Susceptibility Mapping using GIS and Machine Learning 2026
B.Tech Project under Prof. Susmita Panda | *Python, scikit-learn, XGBoost, Google Earth Engine* Live app | GitHub

- Built a landslide-risk model on 4,431 events and 14 satellite-derived factors, comparing kNN, SVM, ANN, Random Forest and XGBoost.
- Showed random splits leak information (XGBoost AUC 0.97 vs 0.85 on held-out 5 km blocks), the same trap as look-ahead bias in a backtest; tuned on spatial cross-validation, and the final SVM scored AUC 0.905 on unseen regions.

Key Factors for Initial Funding of Indian Startups 2025
Research paper with Somya Jangir under Dr. Sanyka Banerjee Report

- Compiled and cleaned data on 585 funded Indian startups (2019–21, USD 1.95B) and built a six-factor framework of what drives early-stage funding.

COMPETITIONS AND TRAINING

Inter IIT Tech Meet 13.0 (Zelta Labs) 2024
Quant Team Member, IIT Jodhpur Predictive Modelling, Time Series

- Built crypto price-direction models for IIT Jodhpur's team; the team ranked 11th overall at the national Inter IIT Tech Meet.
- Engineered features from financial time series and applied statistical tests to check that model signals were not noise.

VC Analyst Case Study (Fin Maverick) *team ranked 4th of 159; picked 3 of 20 startups for a USD 10M fund* 2026
Quantitative Finance Cohort'24 *Quant Delegate:* stochastic calculus, options pricing, volatility models (Certificate) 2024

ACHIEVEMENTS

- **IOQM 2020:** top 300 in Odisha (Certificate). **JEE Main 2023:** top 1% of 1.3 million candidates. **CBSE Class X (2020):** 92%.
- **WorldQuant Challenge 2024:** Bronze level (Certificate). **IICPC 2024:** Competitive Programming Summer Camp (Certificate).

TECHNICAL SKILLS

- **Probability and Statistics:** conditional probability, expectation, hypothesis tests, regression, time series (HAR, Newey-West)
- **Derivatives:** Black-Scholes, Greeks, put-call parity, delta hedging, implied vs realised volatility
- **Programming:** Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost), SQL, C++

POSITIONS OF RESPONSIBILITY

Assistant Head of Finance, Prometeo'25 *budgeting for events and artists, about INR 15 lakh of an INR 40 lakh+ budget* 2025
Head of Media, Prometeo'26 *led 25+ juniors; several reels crossed 500k views* 2025 – 2026
Cultural Captain, Inter IIT Cultural Meet 7.0 and 8.0 *led a 25+ member contingent; top-10 finishes* 2024 – 2025